EXECUTIVE BRIEFING · MEDICINE & LAW

It sounds right. That’s the problem.

Why confident professionals are among the most exposed users of large language models — explained through a 54-item failure taxonomy and real cases.

[Presenter name] · [Organisation] · [Date]

How a language model actually answers

1

Trained on text

Learns patterns of language from vast amounts of text. It stores no list of verified facts.

2

Reads your prompt

Your wording, assumptions and tone become the text it continues.

3

Writes word by word

Each next word is chosen because it is plausible, not because it was checked.

4

Optional: looks things up

Some tools first search the web or your files. The answer is still generated.

The missing step

By default, nothing in this chain checks the answer against reality. Any checking is added by the product — or by you.

Five assumptions that don’t hold

What we assume	What actually happens	What it means for you
“It looks the facts up.”	It writes the most plausible next words. Nothing is looked up unless the tool explicitly searches a source.	A fluent answer can be entirely invented.
“It’s a smarter search engine.”	Search returns existing pages you can open. A language model produces new text that may have no source at all.	Always ask: where exactly is this from?
“It will say if it doesn’t know.”	Training and testing often reward a confident guess over “I can’t tell from this.”	No warning is not the same as certainty.
“Same question, same answer.”	Wording, framing and pushback change the answer, even when no new facts are added.	One good answer proves nothing about the next.
“A citation means it read the source.”	References can be invented, garbled, or real but not supporting the claim.	Check every citation in the original.

THE CORE MECHANISM

Models are rewarded for guessing

Most tests used to train and rank models score answers like a multiple-choice exam with no penalty for wrong answers:

Right answer	Wrong answer	“I don’t know”
1 point	0	0

A guess can only gain points; admitting uncertainty never can. When you demand a definite answer, you reinforce the same incentive.

Expertise does not protect you

Medicine · Dratsch et al., *Radiology* 2023 · Cologne

82% → 46%

Accuracy of experienced radiologists rating mammograms when the AI suggestion was right versus wrong. Inexperienced readers fell from about 80% to under 20%.

Law · Magesh et al., *J. Empirical Legal Studies* 2025 · Stanford

17–33%

Share of answers from leading **specialist** legal research tools (Lexis+ AI, Westlaw AI) that were incorrect or cited sources that did not support them.

Experts follow confident machine output too, and professional-grade tools still err. The danger is not ignorance; it is **trust in your own field**.

The taxonomy: 54 failure modes, four families

20	**Wrong or invented content**	False facts, made-up citations and quotes, and words in transcripts that nobody said.
20	**Conversations that drift**	Answers get worse over a long exchange, bending toward your view or toward its own earlier mistakes.
12	**Errors in tools that search your files**	Systems sold as “working from your documents” can misread, miss or mix up files before answering.
2	**Accurate copy, wrong fact**	Repeating a document faithfully is not the same as being true, or as sticking only to that document.

Key finding: for almost all 54, the taxonomy names a user habit that makes the failure more likely.

In court, the cost is money, cases and careers

2,200+

Court decisions worldwide flagging AI-fabricated content (tracker, Oct 2026). Many filers were self-represented, not lawyers

$110K

Couvrette v. Wisnovsky (D. Or.): ≈$15K fine plus ≈$94K opposing fees — and the client's case dismissed with prejudice

Careers

Lawyers suspended from practice (Colorado 2023; Nebraska, interim, 2026); a US federal prosecutor fired (2026)

The visible part only

Fines are the smallest cost. Lost cases, client claims and published judgments naming the lawyer are not in any tally.

Beyond fines: liability for what the AI said

Tribunal decision · *Moffatt v. Air Canada* · 2024

You own your chatbot's words

The bot invented a refund rule. The airline argued the bot was a "separate entity" — rejected; it had to pay.

Pending lawsuit · *Wolf River Electric v. Google*

Defamation by AI summary

An AI search summary invented a state lawsuit against the firm, which puts losses at $110–210M. Google denies liability.

Settlements · Character.AI and Google · Jan 2026

Chatbot-harm suits settled

Families, including in a wrongful-death case, alleged chatbots reinforced teenagers' self-harm. Terms undisclosed; no admission.

Contracts · Deloitte · Australia and Canada · 2025

Expert reports, fake sources

A$440K report partly refunded; a C$1.6M health report's citations under review, no refund reported.

Firms that deploy AI are being held to its output — and claims against AI vendors are now being tested.

Germany: it has reached our courts

AG Köln · 312 F 130/25 · 2 July 2025

A certified family-law specialist

Cited a Münchener Kommentar entry under the wrong author with a paragraph that does not exist, a monograph nobody could find, and journal references pointing to unrelated decisions.

Court: explicit warning, citing § 43a Abs. 3 BRAO.

KG Berlin · 17 WF 144/25 · 20 Nov 2025

A complaint built on a phantom BGH ruling

Cited BGH “XII ZB 183/07, FamRZ 2008, 137”: no such decision exists, and that journal page holds an unrelated ruling. Two further BGH citations did not fit.

Court: admonished the lawyer; AI-assisted filings must be checked (§ 43 BRAO).

No German fines yet — but both rulings are published, name the professional-conduct rules, and are read by the profession.

Europe: regulators are acting, courts are testing

€250 M

Google, France, 2024. Breach of commitments to news publishers, including training Bard on their content unannounced

€15M

OpenAI, Italy. GDPR fine (Dec 2024) annulled by a Rome court (Mar 2026) on jurisdiction; merits never reviewed

Art. 5(1)(d)

noyb v. OpenAI, Norway, 2025, pending. ChatGPT invented that a real man murdered his children; argued as breach of GDPR accuracy

2027–28

EU AI Act. High-risk duties now Dec 2027 (Annex III) and Aug 2028 (Annex I, e.g. medical devices)

What applies today: GDPR, professional law and civil liability — and an open question whether a hallucination about a **patient or client** is inaccurate personal data.

In the clinic, it is already in the record

~7M

Medical visits transcribed by one Whisper-based scribe tool, used by 30,000+ clinicians (company estimate, 2024)

38%

Of invented passages judged harmful in a study of the underlying Whisper model (about 1% of transcripts had them)

0

Original recordings kept by that vendor to check against — audio deleted for data safety (as of Oct 2024)

Why it matters

Invented words can enter the chart from silence or room noise (H20). Once the audio is gone, only the clinician’s sign-off stands between the error and the record.

Expert habits are the trigger

What experts naturally type	What the research links it to	Row
“Give me five studies supporting this.”	References that don’t exist, or real ones that don’t support the claim	H05, H07
“Just the number — yes or no.”	A precise figure or verdict the inputs can’t support	H18
“Are you sure? I think that’s wrong.”	A correct answer abandoned without new evidence	D02, D01
“Quote the exact wording.”	Paraphrase presented as an exact quotation	H08
“Shorter. Drop the caveats.”	Negations, exceptions and conditions silently lost	D10
“Did you check all of it?”	Reports of checks that never happened	H16

For lawyers: where it bites

H04 · H05

Invented authority

Cases, provisions or articles that do not exist, in flawless citation format

H06

Corrupted citations

A real case with the wrong year, reporter, court or page

H07

Citation–claim mismatch

A real authority attached to a proposition it does not hold

H08

Fabricated quotation

Reconstructed wording presented as verbatim statute or judgment text

H03 · R11

Superseded or foreign law

Old versions presented as current; jurisdictions blended without labels

D16

Hidden instructions

Text inside a third party’s document can redirect the task

For doctors: where it bites

H18

False precision

An exact probability or diagnosis produced from incomplete inputs

H09

Invented causation

A convincing mechanism for a conclusion the evidence does not establish

D18

Care-omission drift

Accommodating a patient's theory until indicated assessment quietly drops out

H20

Phantom speech

Transcripts with words nobody said, generated from silence or room noise

H19

Seeing what you named

Image descriptions that report an object because the question mentioned it

R01

Misread handwriting

Names, numbers, addresses and negations corrupted when scans are digitised

EXERCISE · 3 MINUTES

Find five problems in this AI answer

Prompt: “Summarise the attached record and tell me if we have a claim. Be concise.”

The discharge summary confirms the patient has no known drug allergies. Current guidance recommends a 10-day course as first-line treatment. In *Hartwell v. St. Brennan Medical Center* (2017), the court held that a 48-hour delay in imaging breaches the standard of care. On these facts, the risk of recurrence is 23.4%. I have reviewed all 112 pages of the record.

Fictional example written for this exercise. The case named does not exist.

Answers on the next slide

Five sentences, five different failures

What it said	What went wrong	Row
“No known drug allergies”	The record said “except penicillin.” The exception was lost when asked to be concise.	D10, H11
“Current guidance recommends…”	Which guidance, which year? Possibly superseded; no date given.	H03, R11
Hartwell v. St. Brennan	Invented authority, correctly formatted and plausible.	H04, H05
“Risk of recurrence is 23.4%”	Precise figure with no inputs that could support it.	H18
“Reviewed all 112 pages”	A claim of work, not evidence of it.	H16

Note the prompt: “tell me if we have a claim” invites a verdict; “be concise” invites dropped caveats.

The longer you talk, the further it drifts

Turn 1 · D04

“Could it be X?”

A possibility is raised. Later turns treat it as fact.

→

Turn 2 · D05

“Given X, what follows?”

It elaborates on its own earlier error.

→

Turn 3 · D01 · D02

“Are you sure?”

It moves toward your view, even when it was right.

→

Turn 4 · D06 · D10

“Tighten it up.”

Revision degrades correct work and strips caveats.

Each turn feels like refinement. The research shows it can be **accumulated error** — and the model’s agreement is not new evidence.

LIVE DEMO · 4 MINUTES

The “Are you sure?” test

1

Ask

A question from your field with a known, checkable answer

2

Push back

“Are you sure? I think that’s wrong.” No new evidence

3

Watch

Does it hold, hedge, or reverse — and how confidently?

4

Reset

New chat, give the source, compare the answers

Behaviour varies by model and version; newer models often resist. Rehearse it, and keep a screenshot ready.

“Connected to our records” is not verification

Vendor systems that search your own documents add twelve failure points of their own — most before the model writes a word.

1 Ingest

R01 Scans and handwriting misread

2 Index

R02 Document missing or stale

R03 Exception split from its rule

3 Search

R04 Query rewritten

R05 Similar outranks relevant

R06 Approximate miss

R07 Filtered or cut off

4 Assemble

R08 Evidence trimmed or compressed

5 Read

R09 Answer present, missed

R10 Only part combined

R11 Old version chosen

6 Answer

R12 Unsupported claim or citation

Blue: system stages, invisible to the user · **Orange:** model stages

Faithful to the file is not the same as true

Faithful · True

The goal

Matches the source, and the source is right.

Faithful · False — C01

Errors copied as fact

A wrong allergy in a referral letter or a mis-cited case in a brief, reproduced accurately — and stated as true.

Unfaithful · True — C02

Correct, but not in the record

Adds true background where you asked for “this file only.”

Unfaithful · False — H11 · H12

Contradicts or embellishes

Reverses a negation, or fills gaps with plausible detail.

A fluent explanation is not evidence

H14

Unfaithful reasoning

The stated rationale can omit what actually drove the answer. A good explanation is not an audit trail.

H16

False completion

“I reviewed all 40 pages” may not be true. Ask to see the work, not the claim of work.

D07

Confidence drift

Expressed certainty can rise or fall without any change in correctness. Tone is not a signal.

How strong is this evidence?

The taxonomy grades each of the 54 by how well the **cause** is established. We should hold ourselves to the standard we ask of the tool.



17 · Tested

Mechanism supported by experiments or training-level evidence

13 · By design

System or classification mechanisms in how pipelines are built

24 · Hypothesis

Proposed or working pathway; the failure is described, the exact cause still to be tested

Several sources are 2025–26 preprints and many results are model-specific. The pattern is the message; any single mechanism may be revised.

Count of the “Mechanism” column labels across all 54 rows of the taxonomy

Six rules for professional use

1

Supply the source

Attach the record or the law. Don’t ask it to remember.

2

Check citations three ways

It exists, the details are right, and it supports the claim.

3

Invite “I can’t tell”

Ask what is missing, not for a verdict or a single number.

4

Challenge with evidence

Not pressure. For a second opinion, open a fresh chat.

5

Protect the caveats

Never shorten away negations, exceptions or conditions.

6

Check the work, not the report

Spot-check transcripts, scans and “I reviewed it.”

Each rule targets taxonomy rows: 1 H05 H06 H08 · 2 H05–H07 R12 · 3 H04 H18 · 4 D01 D02 D05 · 5 D10 · 6 H16 H20 R01

Four decisions for leadership

Approved uses	Drafting and summarising with sources supplied. Not unsupervised diagnosis, advice or filings.
Named sign-off	A professional verifies every AI-touched citation, dose, figure and quote before it leaves.
Vendor evidence	Test results for every stage; source audio and logs kept for a defined, lawful review period.
Training and near-miss log	Teach the trigger habits. Log incidents by taxonomy code to see where risk concentrates.

Let it draft.
Never let it decide.

Treat every output as a confident first draft from someone who has never seen the case — and check it the way you would check that person.

Questions and discussion

[Presenter name] · [Contact] · Appendix: sources follow

Sources (1 of 3): research cited in the taxonomy

- Barnett et al., CAIN 2024 — RAG engineering failure points
- Chen et al., 2026 preprint — attention-head interventions in sycophancy
- Scientific Reports 2023 — fabricated citations generated by ChatGPT
- Gao et al., EMNLP 2023 — ALCE citation correctness
- Harshavardhan, 2026 preprint — self-anchoring calibration
- Hsieh et al., 2024 — positional attention calibration
- Huang et al., 2024 — intrinsic self-correction
- Huang et al., TOIS 2025 — factuality and faithfulness taxonomy
- Kalai et al., 2025 — Why Language Models Hallucinate
- Laban et al., 2024 — FlipFlop experiment
- Laban et al., 2025 / ICLR 2026 — LLMs get lost in multi-turn conversation
- Li et al., COLM 2024 — instruction instability and persona drift
- Li et al., EMNLP 2023 — POPE object hallucination
- Liu et al., 2023 — Lost in the Middle
- Lu et al. / Anthropic, 2026 — Assistant Axis
- Mallen et al., ACL 2023 — parametric vs. retrieved knowledge

Full links in the taxonomy file (Sources column)

Sources (2 of 3): research cited in the taxonomy

- Mao et al., 2026 preprint — false accusations and over-correction
- Maynez et al., ACL 2020 — faithfulness and factuality in summaries
- McKinnon / Google Research, 2025 — long-context retrieval quality
- Microsoft Learn — handwritten and printed document OCR
- Microsoft Learn — vector search algorithms and ranking
- NIST AI 600-1, 2024 — generative AI risk profile
- OpenAI, May 2025 — withdrawn GPT-4o sycophancy update
- Russinovich et al., USENIX Security 2025 — Crescendo
- Sharma et al., ICLR 2024 — sycophancy and preference optimisation
- Smyth et al., 2026 preprint — OverclaimBench
- Sun et al., ACL Industry 2026 — OCR robustness in RAG
- Turpin et al., NeurIPS 2023 — unfaithful chain of thought
- Wallace et al., 2024 — instruction hierarchy
- Wang et al., Interspeech 2025 — Calm-Whisper
- Wu et al., ICLR 2025 — LongMemEval
- Zhang et al., ICML 2024 — hallucination snowballing

Full links in the taxonomy file (Sources column)

Sources (3 of 4): cases and liability

- Charlotin, AI Hallucination Cases database — 2,202 entries as of 19 Oct 2026 (via HAQQ)
- EDRM / ComplexDiscovery, Apr 2026 — “The AI Sanction Wave”: at least $145K in Q1 2026 US penalties
- David Kemp, 2026 — “Citation Sanctions in Q1 2026” (Nebraska interim suspension, DOJ dismissal)
- ABA Journal, 17 Apr 2026 — *Couvrette v. Wisnovsky*, D. Or.: $110K in penalties, case dismissed
- ABA Litigation News, 2026 — “Fake Cases, Real Sanctions” (*Whiting v. City of Athens*, 6th Cir.)
- Colorado Politics, 2 Dec 2023 — Colorado attorney suspended over ChatGPT-fabricated cases
- *Mata v. Avianca, Inc.*, S.D.N.Y. 2023 — $5,000 sanction
- *Moffatt v. Air Canada*, 2024 BCCRT 149 (Law360 Canada)
- CO/AI and GovTech, 2025 — *Wolf River Electric v. Google* (AI Overview defamation claim)
- Fortune, 8 Jan 2026 — Character.AI and Google settle chatbot-harm lawsuits
- TechRadar, 7 Oct 2025; SAN, 27 Nov 2025 — Deloitte reports with fabricated citations
- beck-aktuell, 8 Jul 2025 — AG Köln, 312 F 130/25 · dejure.org — KG Berlin, 17 WF 144/25

Sources (4 of 4): studies, regulation and guidance

- Kalai, Nachum, Vempala, Zhang, 2025 — “Why Language Models Hallucinate,” arXiv 2509.04664 (preprint)
- Dratsch et al., *Radiology* 2023 — automation bias in mammography (RSNA News, 2 May 2023)
- Magesh et al., *J. Empirical Legal Studies* 22 (2025) 216–242 — reliability of AI legal research tools
- Koenecke, Sloane et al., FAccT 2024 — “Careless Whisper: Speech-to-Text Hallucination Harms”
- AP via Fortune, 26 Oct 2024 — Whisper-based transcription in hospitals (Nabla figures)
- CNBC, 20 Mar 2024 — Autorité de la concurrence fines Google €250M
- Lewis Silkin, 14 Jan 2025; PPC Land, 2026 — Garante €15M OpenAI fine and its annulment (Trib. Roma, 18 Mar 2026)
- noyb, 20 Mar 2025 — complaint to Datatilsynet under GDPR Art. 5(1)(d)
- Orrick, Jul 2026 — EU AI Act Digital Omnibus: revised high-risk deadlines
- ABA Formal Opinion 512 (2024); NIST AI 600-1 (2024)
- LLM hallucination & drift taxonomy v1 — 54 rows, the basis of this briefing

Incident and regulatory status checked 24 October 2026; trackers and proceedings change continuously